

Case no. 1/ Presentation

- A 60-year-old man with a 30-year history of alcohol abuse and T2DM of 26 years, presented to emergency department because of hematemesis for the past day.
- O/E, he has ascites, mild jaundice, enlarged spleen, gynecomastia, telangiectasias, and testicular atrophy.
Rectal examination positive for hemorrhoids.

- **Laboratory studies:**

- Total protein = 5.8 g/dL; albumin = 3.4 g/dL;
- Total bilirubin= 5.4 mg/dL; direct bilirubin= 3 mg/dL;
- AST= 137 U/L; ALT= 108 U/L;
- Alkaline phosphatase= 181 U/L;
- Prothrombin time= 20 seconds;
- Hematocrit= 25%, WBC= $3.1 \times 10^9/L$, Plt= $90 \times 10^9/L$

Total protein.	= 6.3 - 7.9 g/dL.
Albumin.	= 3.5 - 5.0 g/dL.
Bilirubin.	= 0.1 - 1.2 mg/dL.
ALT.	= 7 - 55 U/L.
AST.	= 8 - 48 U/L.
ALP.	= 40 - 129 U/L.
GGT.	= 8 - 61 U/L.
LD.	= 122 - 222 U/L.
PT.	= 9.4 - 12.5 seconds.

Case no. 1/ Questions

What is the most likely diagnosis?

What is the cause of his condition?

What is the cause of hematemesis, what is the pathogenesis of this condition?

What other findings are also related to portosystemic shunt?

Case no. 1/ Questions

What is the cause of gynecomastia, telangiectasias, and testicular atrophy?

Hormonal imbalances resulting from liver dysfunction.

Decrease estrogen metabolism leading to hyperestrogenemia.

Case no. 1/ Questions

Patient has ascites.

The ascitic fluid was aspirated, what will be the type of the fluid, how can you confirm it by biochemical testing?

Transudative

Protein content

What is the pathogenesis of ascites in liver cirrhosis?

Case no. 1/ Questions

Patient has abnormal hematologic findings.

What is the pathogenesis of these findings?

Case no. 1/ Questions

Abdominal US is done, what are the likely findings to be seen?

A liver biopsy is taken, what are the histologic findings?

Case no. 1/ Questions

- The patient was admitted to hospital, on the 6th day, his condition deteriorate, as he become progressively irritable and confused, physical examination showed Asterixis.

Case no. 1/ Questions

Which of the following is most closely associated with the development of hepatic encephalopathy?

- A) Ascites.
- B) Hypersplenism.
- C) Portosystemic shunts.
- D) Hyperkalemia.
- E) Hyperestrogenemia.

Case no. 1/ Questions

- What are the possible participate factors for the development of such condition?

Case no. 1/ Questions

How does GIT bleeding
precipitates hepatic
encephalopathy?

Case no. 2/ Presentation:

A 64 years retired nurse female, presented with increasing fatigue, loss of appetite, unintentional weight loss of about 12kg over the past three months, and worsening right upper quadrant abdominal pain.

PMH:

Positive for hepatitis B, diagnosed at age 43 years.

History of skin melanoma, excised at age 59 years.

O/E:

Appears thin, weigh 46kg.

Tenderness and a palpable mass in the right upper quadrant.

Moderate ascites was present.

Case no. 2/ Questions:

What are the possible differential diagnoses for her condition?

What investigations do you like to order?

- Liver Function Tests:

- AST (210 U/L) and ALT (260 U/L)
- Alkaline phosphatase (250 U/L)
- Total bilirubin (3.3 mg/dL)
- Albumin (2 g/dL)
- Prothrombin time (20 seconds)

- Tumor Markers:

- Alpha-fetoprotein (AFP) level (1500 ng/mL)

Total protein.	= 6.3 - 7.9 g/dL.
Albumin.	= 3.5 - 5.0 g/dL.
Bilirubin.	= 0.1 - 1.2 mg/dL.
ALT.	= 7 - 55 U/L.
AST.	= 8 - 48 U/L.
ALP.	= 40 - 129 U/L.
GGT.	= 8 - 61 U/L.
LD.	= 122 - 222 U/L.
PT.	= 9.4 - 12.5 seconds.
AFP	= 0 - 40 ng/mL

Case no. 2/ Questions:

The US report states

“ the liver is shrunken in size, have nodular contour all over but contain two larger and poorly defined masses in the right lobe measured 3cm and 6cm across, these masses appear darker/ hypoechoic and more vascular than rest of the surrounding much smaller nodules.

Several enlarges hilar lymph nodes were seen.

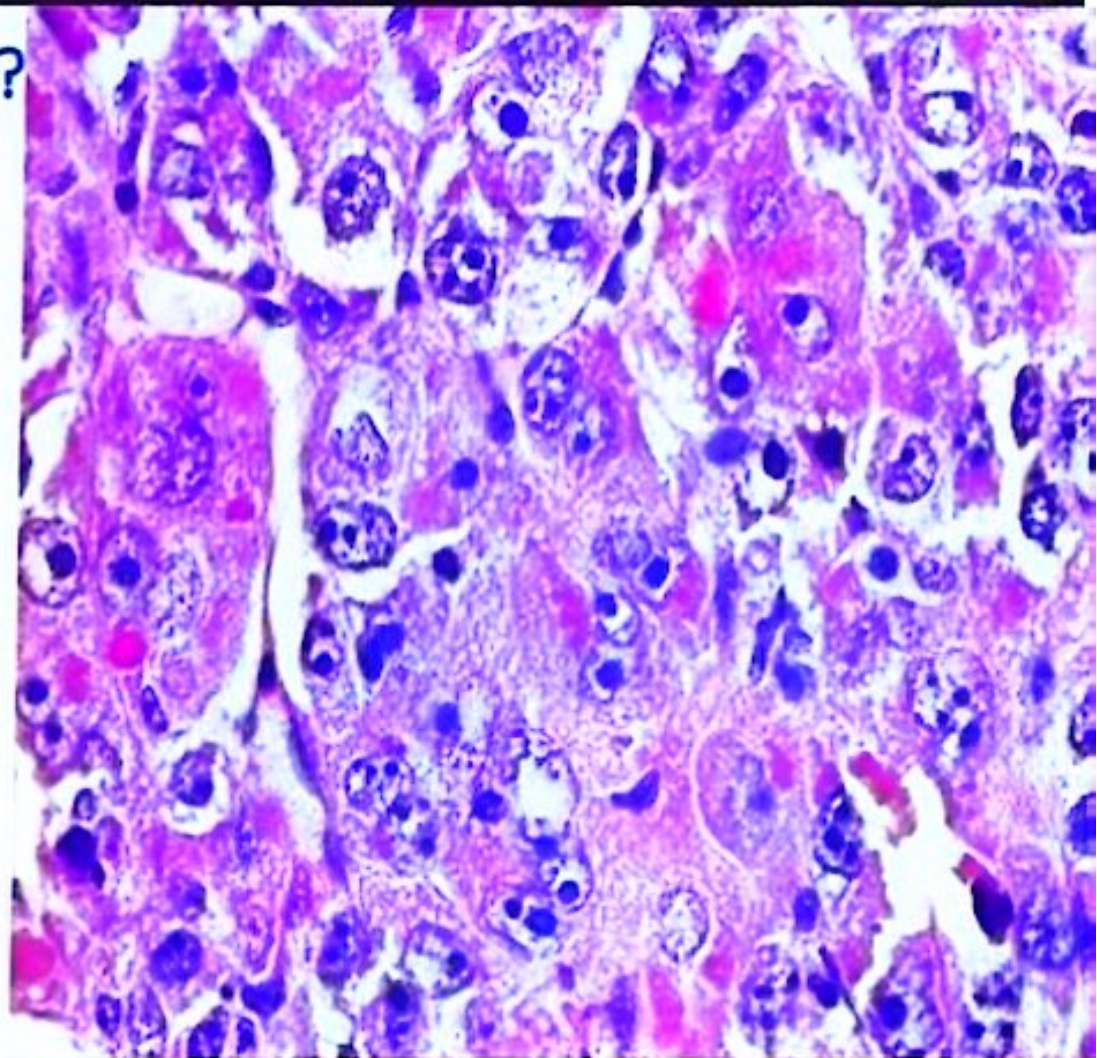
Moderate ascites was present.”.

What is the likely diagnosis by now?

Case no. 2/ Questions:

How do you confirm your diagnosis?

This is the picture of the biopsy,
what are the histologic findings?



Case no. 2/ Questions:

What is the most likely cause for her condition?

What is the likely type of ascites, how you confirm it?